

Jakub Kisielewski

07397 083677 | jakub.kisielewski@warwick.ac.uk | github.com/Jakub-Kisielewski | linkedin.com/in/jakub-kisielewski-30360b27b/
EDUCATION

University of Warwick

BSc (Hons) in Computer Science (Expected First)

Coleg Gwent

A-levels in Mathematics, Further Mathematics, Computer Science and Physics

Coventry, England

September 2024 - June 2027

Crosskeys, Wales

September 2022 - June 2024

TECHNICAL SKILLS

Languages: Java, C/C++, Python, SQL (Postgres), GDScript, Angular (HTML, CSS, JS, TS)

Developer Tools: Git, GitHub/GitLab, Maven, Godot, CMake, Flyway, Lombok, Valgrind, Neovim

Concepts: Operating Systems & Networks, Data-Level Parallelism, Low Latency Systems, Memory Systems

EXPERIENCE

Software Engineering Intern @ BGC | *Java, Spring, Angular, PSQL, Flyway, Mockito* July – September 2026

- Contributed to **30+** user internal tool, enhancing DB service usability with REST API, auditing, and bespoke UI
- Built understanding of existing code through probing edge cases/tracing downstreams to inform REST API design
- Completed research to expose existing auditing solution **JaVers**, now used **across Post Trade** projects at BGC
- Automated manual workflow with an **~85%** (15 → 2min) time save by replacing ad-hoc SQL with an intuitive UI

PROJECTS

Financial Exchange Simulation | *Java, Maven, Netty, SQL (Postgres)* July – August 2026

- Designed **O(1)** orderbook operations using **RB Trees**, hashmaps & doubly linked lists to simulate 100s of orders/s
- Minimised JVM Garbage Collection, reducing **Stop-The-World** pauses, ensuring **microsecond** granularity
- Simulated **100s** of traders & market makers over **TCP**, using the **FIX** protocol for realistic communication

Pixel Painter: Race | *C++, RayLib, CMake* June 2026

- Constructed a real-time, **dual-client** game with **no game engine** and Raylib for hardware-accelerated rendering
- Engineered custom, low-latency networking with socket programming to sync **625** data points bidirectionally
- Decoupled network layer from rendering with **async** multithreading and byte-level serialisation to reduce payload

Airport Traffic Model | *Java, JavaFX, JUnit* January – March 2026

- Collaborated in team of **6** to build **real-time** simulation, analysing aircraft throughput in dynamic airport layouts
- Designed **multithreaded** architecture, decoupling sim engine from UI ensuring **stutter-free** real-time rendering
- Implemented dynamic **data decimation** and **throttling** of JavaFX chart updates, preventing UI lag in large sims

Crowd Flow | *Python, Streamlit, Numba, NumPy, Librosa* February 2026

- Implemented a predictive framework to help event organisers plan barrier placements to **protect** concert attendees
- Built a custom Verlet Integration physics engine utilising **JIT compilation** to simulate **2000+** agents at 60FPS
- Modelled realistic & dynamic crowds, with waveform analysis of audio to modulate agent behaviour in **real-time**
- Reduced crowd pressure by **~60%** using a **Monte Carlo Search** engine to generate optimal barrier placement

Maze Runner | *GDScript, Godot, Git* October – December 2025

- Developed and released a 2D rogue-like game in a **3-person** team, utilising GDDs to align technical milestones
- Defined procedural maze creation with randomised DFS + heuristics, reducing brute-force generation overhead
- Integrated audio management system, orchestrating over **50+** spatial SFX and music tracks across game states

Packet Sniffer | *C, POSIX, UNIX APIs* November – December 2025

- Created a **multi-threaded** program to detect and prevent multiple network attacks at high-throughput interfaces
- Established **thread pool** architecture with hardware dependent thread count, ensuring synchronisation with locks
- Utilised **libpcap** to hook into raw network sockets, manually verifying **IPv4** packets to identify malicious activity
- Verified **100%** packet capture accuracy during high traffic periods by dynamically reallocating memory

Text Editor | *C, Linux Kernel, GCC* August – September 2025

- Constructed a lightweight terminal text editor featuring syntax highlighting, filetype detection, and file persistence
- Engineered low-level, real-time input handling using UNIX system calls (read, write, ioctl, ftruncate, open, close)
- Optimised speed by building efficient data structures and buffered screen rendering leading to **22μs boot time**

VOLUNTEERING/LEADERSHIP

University of Warwick Computing Society - Treasurer March 2025 - Present

- Elected treasurer of the biggest tech society at Warwick, beating **4** other candidates and working in a team of **30**
- Accumulated **£8500** in sponsorship money to ensure the smooth running of all events for **400+** members

WHACK Hackathon - Co-Lead Organiser 25/26 March 2025 - Present

- Raised **£9000** in sponsorship money to successfully run a **200** attendee hackathon on Warwick campus in 2025
- Raised **£16000** in sponsorship money to continue the success of WHACK into 2026
- Presently managing **20** organisers to double event size for 2026 and ensure smooth future organising of WHACK